

Chapter 5

Analysis of Open Systems / Control Volumes

A **control mass** is a defined *piece of matter*. The matter may move around, and the boundaries of the control mass follow the matter wherever it goes. By definition, no matter can enter or leave a control mass, and therefore the mass of a control mass is always the same. In a control mass energy analysis, the only forms of energy transfer across the boundary are work and heat.

A **control volume** is defined as a *region of space*. Matter may enter or leave a control volume, and the matter in a control volume is whatever happens to be in it at any moment. For example, a control volume can be defined as the space around an aircraft jet engine, or the space inside a balloon. The control volume boundary may move through space (as with the jet engine), and may change shape and volume (as with a balloon). The important thing is that the boundary must be well-defined and identifiable at all times. In a control volume energy balance, the possible energy transfers across the boundary include heat, work, and energy transfer associated with mass flow across the boundary. The primary point of this chapter is to derive the correct expressions for this mass-associated energy transfer.

5.1 Conservation of Mass for a Control Volume

For each of the extensive properties - mass, energy, entropy and exergy, the control volume form of the property balance can be obtained by transforming the corresponding closed system form. Let us consider this for mass, recalling that the mass of a closed system is constant. **Figure 5.1** show a system consisting of a fixed quantity of matter m that occupies different regions at time t and a later time $t + \Delta t$. The mass under consideration is shown in color on the figures. At time t , the mass is the sum $m = m_{cv}(t) + m_i$, where $m_{cv}(t)$ is the mass contained within the control volume, and m_i is the mass within the small region labeled i adjacent to the control volume. Let us study the fixed quantity of matter m as time elapses. In a time interval Δt all mass in region i crosses the control volume boundary, while some of the mass, call it m_e , initially contained within the control volume, exits to fill the region labeled e adjacent to the control volume. Although the mass in regions i and e as well in the control volume differs from time t to $t + \Delta t$, the *total* amount of mass is constant. Accordingly,

$$m_{cv}(t) + m_i = m_{cv}(t + \Delta t) + m_e \quad (5.1)$$

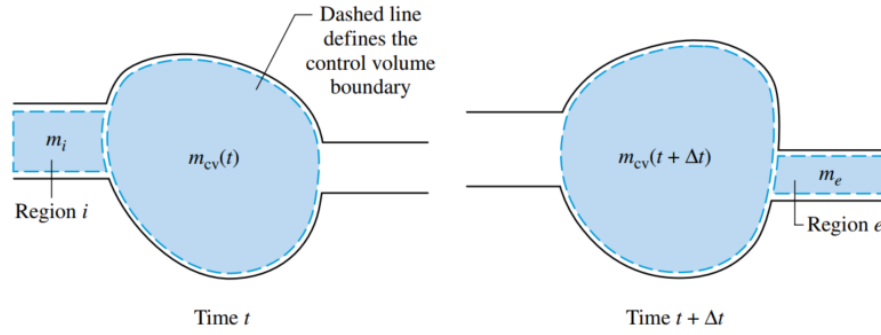


Figure 5.1: Schematic for deriving mass balance relation for control volume.

Upon rearrangement it becomes,

$$m_{cv}(t + \Delta t) - m_{cv}(t) = m_i - m_e \quad (5.2)$$

The equation can be expressed on a time rate basis. First divide by Δt to obtain:

$$\frac{m_{cv}(t + \Delta t) - m_{cv}(t)}{\Delta t} = \frac{m_i}{\Delta t} - \frac{m_e}{\Delta t} \quad (5.3)$$

Then, in the limit as Δt goes to zero, the equation becomes

$$\frac{dm_{cv}}{dt} = \dot{m}_i - \dot{m}_e \quad (5.4)$$

where $\frac{dm_{cv}}{dt}$ denotes the time rate of change of mass within the control volume, and $\dot{m}_i$ and $\dot{m}_e$ are the inlet and exit mass flow rates, respectively, all at time t .

The relation derived in [Equation \(5.4\)](#) applies to control volumes with one inlet and one exit. If there are multiple inlet and exit ports, the following relation could be easily derived:

$$\frac{dm_{cv}}{dt} = \sum_i \dot{m}_i - \sum_e \dot{m}_e \quad (5.5)$$

Here, $\dot{m} = \rho A V = A V / v$, due to which [Equation \(5.5\)](#) can be alternatively written as:

$$\frac{dm_{cv}}{dt} = \sum_i \frac{A_i V_i}{v_i} - \sum_e \frac{A_e V_e}{v_e} \quad (5.6)$$

The expressions for mass balance derived so far assume a **one-dimensional flow**, i.e., the flow is normal to the boundary at locations where mass enters or exits the control volume, and all intensive properties, including velocity and density, are uniform over each inlet and exit area through which matter flows. If the intensive properties, however, vary over position, the **integral form** of the mass-balance equation will be more useful. The total mass contained within the control volume at an instant t can be related to the local density as follow:

$$m_{cv}(t) = \int_V \rho dV \quad (5.7)$$

The mass flow rate over inlet and exit ports could be written in the integral form as $\dot{m} = \int_A \rho V_n dA$. The mass balance [Equation \(5.5\)](#) could be written in the integral form as

$$\frac{d}{dt} \int_V \rho dV = \sum_i \left(\int_A \rho V_n dA \right)_i - \sum_e \left(\int_A \rho V_n dA \right)_e \quad (5.8)$$

Here, V_n represents the normal velocity component at the inlet and exit ports, and ρV_n represents the **mass flux** - time rate of mass flow per unit of area. The two terms on the right could be collapsed into one by writing the integral form of mass balance equation as:

$$\frac{d}{dt} \int_V \rho dV = - \sum_{i,e} \left(\int_A \rho (\vec{V} \cdot \hat{n}) dA \right)_{i,e} \quad (5.9)$$

Here, $\hat{n}$ represents the outward normal unit vector at each infinitesimal area dA over inlet and outlet ports. Thus, $\vec{V} \cdot \hat{n}$ represents the normal velocity, which is positive for outlet ports and negative for inlet ports. The normal $\hat{n}$ points from the control volume outwards.

Example Problem 01

Water flows into the top of an open barrel at a constant mass flow rate of $\dot{m}_i = C_1$. Water exits through a pipe near the base with a mass flow rate proportional to the height of liquid inside: $\dot{m}_e = C_2 L$, where L is the instantaneous liquid height. The area of the base is A_b , and density of water is ρ_w . If the barrel is initially empty, find the variation of liquid height with time.

Solution

Let us first write the rate of mass balance equation - it's one dimensional form, since we are assuming that the flow is *one-dimensional* - intensive properties are uniform along inlet and outlet ports; velocity is normal to the inlet and outlet ports.

$$\frac{dm_{cv}}{dt} = \dot{m}_i - \dot{m}_e \quad (5.10)$$

This can be re-written as:

$$\frac{d}{dt} (\rho_w A_b L) = C_1 - C_2 L \quad (5.11)$$

which could be further simplified to:

$$\frac{dL}{dt} + \left(\frac{C_2}{\rho_w A_b} \right) L = \frac{C_1}{\rho_w A_b} \quad (5.12)$$

This is a linear ODE of first order, which could be solved with the help of an integrating factor, $\exp(C_2 t / \rho_w A_b)$, to arrive at a solution:

$$L(t) = \left(\frac{C_1}{C_2} \right) \left(1 - \exp \left(-\frac{C_2 t}{\rho_w A_b} \right) \right) \quad (5.13)$$

In deriving the solution, we have used the fact that at $t = 0$, $L = 0$.

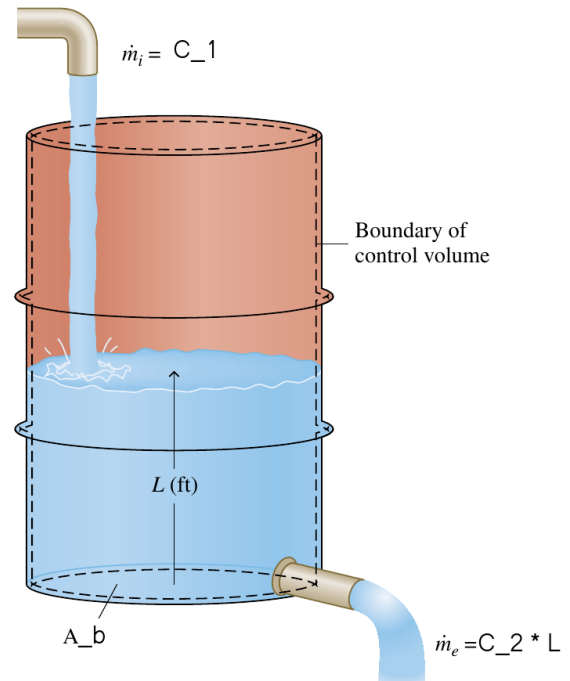


Figure 5.2: Schematic for Example 01 - Transient emptying of a water tank.

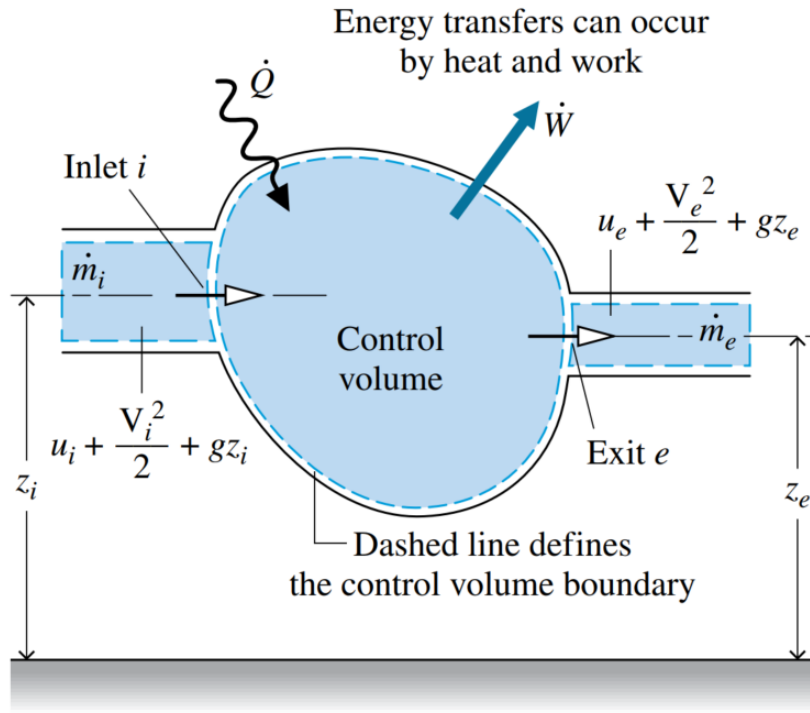


Figure 5.3: Schematic used to develop energy balance expression for a control volume.

5.2 Conservation of Energy for a Control Volume

We have seen in the previous chapter that the rate of energy balance equation for a control mass could be written as:

$$\frac{dE}{dt} = \dot{Q} - \dot{W}$$

where, $\dot{Q}$ is the heat input to the control mass from surroundings and $\dot{W}$ is the work done by the system. Consider a control volume as shown in [Figure 5.3](#). Here the energy is transferred not only as heat and work, as in the control mass, but also due to the mass transfer taking place at the boundaries. In [Figure 5.3](#), we have considered a control volume with just one inlet and one exit port. The energy is transported by the mass flow at the inlet and exit ports. The energy equation for the control volume then becomes:

$$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W} + \dot{m} \left(u_i + \frac{V_i^2}{2} + gz_i \right) - \dot{m} \left(u_e + \frac{V_e^2}{2} + gz_e \right) \quad (5.14)$$

where E_{cv} denotes the energy of the control volume at time t . The terms $\dot{Q}$ and $\dot{W}$, account, respectively, for the net rate of energy transfer by heat and work across the boundary of the control volume at t . It is convenient to separate the **work term** $\dot{W}$ into *two contributions* - One contribution is the work associated with the fluid pressure as mass is introduced at the inlets and removed at exits. The other contribution, denoted by $\dot{W}_{cv}$, includes *all other* work effects, such as those associated with rotating shafts, displacement of the boundary (e.g., due to expansion of control volume), and electrical effects.

Consider matter removed from the system at the exit station e . Since matter is getting displaced, work is done by the system due to its pressure. The pressure of the system always acts outwards. Since flow of matter is also outwards at the exit station e , work done by the system will be positive and the rate of work done will have a magnitude $p_e A_e V_e$, where p_e is the pressure, A_e is the area, and V_e is the normal velocity at exit e , respectively. The expression for rate of work can be derived from

the fact that rate of work done is simply the scalar dot product of force and matter velocity. Using similar arguments, we can arrive at the work done by the system at the inlet station i : $-p_i A_i V_i$. The negative sign is due to the fact that pressure p_i acts outwards and matter is flowing into the system.

With these considerations, the work term $\dot{W}$ of the energy rate equation, Equation (5.14), can be written as:

$$\dot{W} = \dot{W}_{cv} + (p_e A_e) V_e - (p_i A_i) V_i \quad (5.15)$$

The equation above can be further simplified by using the relation $\dot{m} = \rho A V$ or $\dot{m} v = A V$. The expression above can be written as -

$$\dot{W} = \dot{W}_{cv} + \dot{m}_e (p_e v_e) - \dot{m}_i (p_i v_i) \quad (5.16)$$

Here, $\dot{m}_i$ and $\dot{m}_e$ are the mass flow rates and v_i and v_e are the specific volumes evaluated at the inlet and exit, respectively. the terms $\dot{m}_i (p_i v_i)$ and $\dot{m}_e (p_e v_e)$ account for the work associated with the pressure at the inlet and exit, respectively. They are commonly referred to as the **flow work**. The term $\dot{W}_{cv}$ accounts for *all other* energy transfers by work across the boundary of the control volume - rotating shafts, displacement of the boundary, and electrical effects.

Substituting Equation (5.16) into Equation (5.14), and collecting all the terms belonging to the inlet and exit into separate expressions, the following form of the control volume energy rate balance results:

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_i \left(u_i + p_i v_i + \frac{V_i^2}{2} + g z_i \right) - \dot{m}_e \left(u_e + p_e v_e + \frac{V_e^2}{2} + g z_e \right) \quad (5.17)$$

The heat term $\dot{Q}$ has been rewritten as $\dot{Q}_{cv}$ to remind that the term represents heat transfer across the boundary of a control volume. The last two terms can be rewritten using the **specific enthalpy** h . With $h = u + p v$, the energy rate balance becomes:

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_i \left(h_i + \frac{V_i^2}{2} + g z_i \right) - \dot{m}_e \left(h_e + \frac{V_e^2}{2} + g z_e \right) \quad (5.18)$$

The appearance of the sum $u + p v$ in the control volume energy equation is the principal reason for introducing enthalpy previously. It is brought in solely as a *convenience*: the algebraic form of the energy rate balance is simplified by the use of enthalpy.

If there are several locations on the boundary through which mass enters or exits. This can be accounted for by introducing summations as in the mass balance. Accordingly, **energy rate balance** is:

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum_i \dot{m}_i \left(h_i + \frac{V_i^2}{2} + g z_i \right) - \sum_e \dot{m}_e \left(h_e + \frac{V_e^2}{2} + g z_e \right) \quad (5.19)$$

Equation (5.19) is a one-dimensional form of the energy rate balance, i.e., the flow is normal to the boundary at locations where mass enters or exits the control volume, and all intensive properties, including velocity and density, are uniform over each inlet and exit area through which matter flows. If the intensive properties vary over position, the **integral form** of the energy rate balance equation will be more useful. The total energy contained within a control volume at time t can be written as:

$$E_{cv}(t) = \int_V \rho e dV = \int_V \rho \left(u + \frac{V^2}{2} + g z \right) dV \quad (5.20)$$

Similarly, the terms accounting for the energy transfers accompanying mass flow and flow work at inlets and exits can be expressed as shown in the following form of the energy rate balance:

$$\frac{d}{dt} \int_V \rho e dV = \dot{Q}_{cv} - \dot{W}_{cv} + \sum_i \left[\int_A \left(h + \frac{V^2}{2} + g z \right) \rho V_n dA \right]_i - \sum_e \left[\int_A \left(h + \frac{V^2}{2} + g z \right) \rho V_n dA \right]_e \quad (5.21)$$

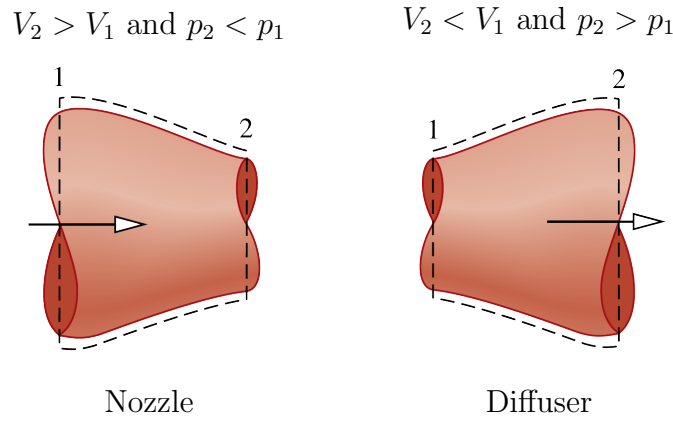


Figure 5.4: Illustration of a nozzle and a diffuser at subsonic conditions.

In situations where $\dot{Q}_{cv} = 0$ and $\dot{W}_{cv} = 0$ in a one-inlet and one-exit systems, we can write Equation (5.27) as:

$$\begin{aligned} h_i + \frac{V_i^2}{2} + g z_i &= h_e + \frac{V_e^2}{2} + g z_e \\ u_i + p_i v_i + \frac{V_i^2}{2} + g z_i &= u_e + p_e v_e + \frac{V_e^2}{2} + g z_e \end{aligned} \quad (5.28)$$

If the matter flowing through the control volume is either an ideal gas or incompressible fluid, $u = u(T)$. Thus if the system and surroundings are at the same temperature and the entire process is isothermal, the internal energy terms appearing on both sides in Equation (5.28) cancel out, and we are left with:

$$p_i v_i + \frac{V_i^2}{2} + g z_i = p_e v_e + \frac{V_e^2}{2} + g z_e \quad (5.29)$$

This familiar looking equation is called the **Bernoulli's equation**. Since it is derived from the steady state energy equation, Bernoulli's equation is a statement of energy balance. The same equation can also be derived from momentum balance equation on a fluid particle, which also makes it a statement of momentum balance.

5.4 Nozzles and Diffusers

We now apply the rate balance equations to some real world devices, including nozzles and diffusers, turbines, compressors and pumps, heat exchangers and throttling devices. These devices are assembled to form complex equipment like power plants and refrigeration units.

A **nozzle** is a flow passage of varying cross-sectional area in which the velocity of a gas or liquid increases in the direction of flow. In a **diffuser**, the gas or liquid decelerates in the direction of the flow.

Figures 5.4 and 5.5 show nozzle and diffuser cross sectional area variation at **subsonic** and **supersonic** conditions. At subsonic conditions, the flow speed of the fluid passing through is less than the speed of sound in that fluid. In supersonic flow, the fluid speed is higher than the speed of sound. In subsonic conditions, the nozzle's cross-sectional area decreases in the direction of flow while a diffuser's cross-sectional area increases in the direction of flow. For supersonic conditions the situation is the opposite as shown in Figure 5.5. Thus to accelerate the fluid from low speed (subsonic) to very high speed

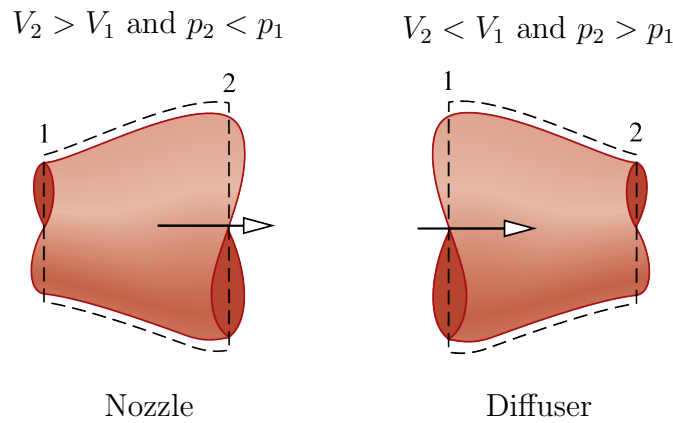
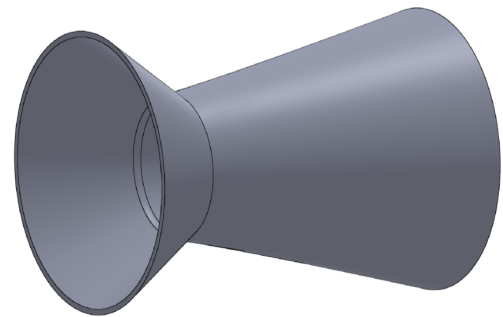


Figure 5.5: Illustration of a nozzle and a diffuser at supersonic conditions.

(supersonic) conditions, one will require special type of nozzles - called the **converging diverging** nozzle as shown in **Figure 5.6**.

The nozzle consists of a converging section where the flow speed increases from subsonic low speeds to sonic speed (speed of sound) at the throat. Throat is the minimum cross-sectional area of a converging diverging nozzle. Beyond the throat, the flow is further accelerated to supersonic speeds in the diverging section. Converging-diverging nozzles find applications in steam turbines, rockets and supersonic jet engines.



In an idealized nozzle, the heat transfer rate $\dot{Q}_{cv}$ and work $\dot{W}_{cv}$ are negligible compared to the changes in enthalpy and the kinetic energy. The change in potential energy can also be neglected in typically available nozzles. Thus, **Equation (5.27)** can be simplified to:

Figure 5.6: A 3D rendered image of a converging diverging nozzle.

$$h_i - h_e + \frac{V_i^2 - V_e^2}{2} = 0 \quad (5.30)$$

Example Problem 02

Steam enters a converging-diverging nozzle operating at steady-state with $p_1 = 40$ bar, $T_1 = 400^\circ$ C, and a velocity of 10 m/s. the steam flows through the nozzle with negligible heat transfer and no significant change in potential energy. At the exit, $p_2 = 15$ bar, and the velocity is 665 m/s. The mass flow rate is 2 kg/s. Determine the exit area of the nozzle, in m^2 .

Solution

Let us first apply the **conservation of mass** equation for steady-state flow through one-inlet one-exit control volume.

$$\frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2} = \dot{m} = 2 \text{ kg/s} \quad (5.31)$$

We need to calculate A_2 at station 2 as required in the question. From **Equation (5.31)**, we require v_2 , to compute A_2 . However, we are only given pressure at that station, which is not sufficient to

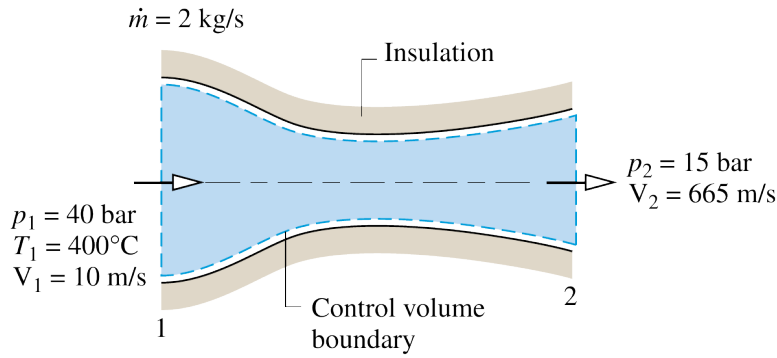


Figure 5.7: Control volume demarcated in the converging diverging nozzle of Example Problem 02.

fix the state of fluid at that station. Fixing the state will enable us to compute all other intensive properties including specific volume v_2 . We will use an additional relation to arrive at one more intensive property, that will *fix* the state at station 2 for us. That additional relation is steady-state **energy rate balance** for one-inlet one-exit systems.

As given in the problem, the heat transfer $\dot{Q}_{cv}$ can be neglected alongwith the change in potential energy. Additionally, $\dot{W}_{cv} = 0$ since there is no shaft-work, no electrical work and no work done due to moving boundaries. We have, for the nozzle,

$$0 = h_1 - h_2 + \frac{V_1^2 - V_2^2}{2}$$

Solving for h_2 ,

$$h_2 = h_1 + \frac{V_1^2 - V_2^2}{2}$$

We require h_1 at station 1. Now, at state 1, two intensive properties - temperature and pressure - are given. These two properties should be sufficient to *fix* the state at station 1, if the steam at station 1 is in a single state of matter: solid, liquid or gas. We will use the phase diagram ($p-T$ diagram) to ascertain which phase the steam is in at station 1. From the saturation table of water, the saturation temperature corresponding to $p_1 = 40$ bar (4000 kPa) is 250.4 °C. However, the temperature at station 1 is 400 °C, which makes it a superheated, actually supercritical, vapor. Thus temperature and pressure specified at station 1 are sufficient to *fix* the system and we need to lookup the superheated vapor tables to read the correct value of h at station 1. Corresponding to 40 bar (4 MPa) and 400 °C, the enthalpy from superheated water table is 3213.6 kJ/kg. Thus h_2 can be calculated as:

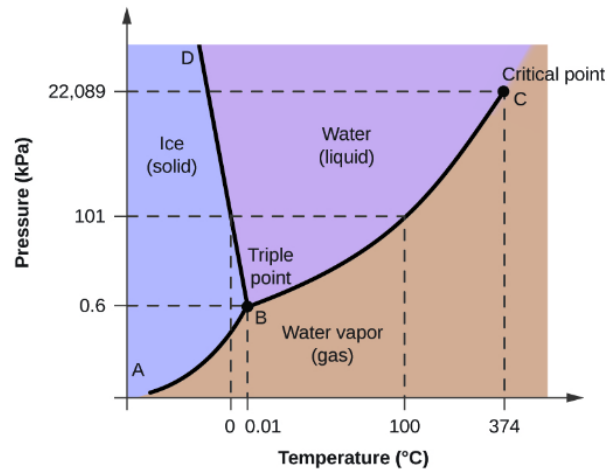


Figure 5.8: Phase diagram of water.

$$\begin{aligned} h_2 &= 3213.6 \times 10^3 \text{ J/kg} + \frac{10^2 - 665^2}{2} \text{ J/kg} \\ &= 2992.5 \text{ kJ/kg} \end{aligned}$$

We now go back to the saturation table at $p_2 = 15$ bar to see if the calculated h_2 lies between h_f and h_g values at saturation. If h_2 is lower than h_f , the water at station 2 is in liquid state. If h_2 is

greater than h_g , water at station 2 is in superheated vapor state, and if h_2 is between h_f and h_g , then the water at station 2 is in saturated state. This is another way of determining the state of matter of a system if temperature and pressure are not provided, e.g., at station 2, and thus phase diagram cannot be used.

The saturation pressure tables provides the values of h_f and h_g at $p_2 = 15$ bar: $h_f = 844.84$ kJ/kg and $h_g = 2792.2$ kJ/kg. Since h_2 is greater than h_g , the water at station 2 is in superheated vapor state. So we look at the superheated vapor table at $p_2 = 15$ bar. We find that at $T = 280^\circ\text{C}$, $h = 2992.7$ kJ/kg, which is very close to h_2 . Thus we can say that T_2 is approximately 280°C .

5.5 Turbines

A turbine is a device in which power is developed as a result of a gas or liquid passing through a set of blades attached to a shaft free to rotate. A schematic of an axial-flow steam or gas turbine is shown in Figure 5.9. Such turbines are widely used for power generation in vapor power plants, gas turbine power plants, and aircraft engines. In these applications, superheated steam or a gas enters the turbine and expands to a lower pressure as power is generated. In schematics of thermal systems, e.g. thermal power plants, the turbine is represented using the same symbol as a subsonic diffuser.

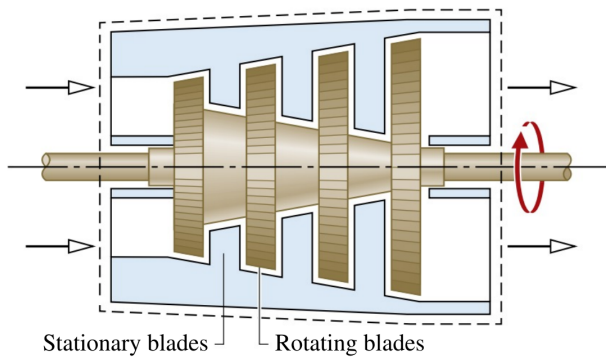


Figure 5.9: Schematic of an axial-flow steam or gas turbine.

In a **hydraulic** turbine coupled to a generator, the water flows from higher to lower elevation through the turbine. The power generated in the turbine shaft is provided to the generator, which converts it to electricity. This type of power is called **hydropower**.

We will now consider an energy analysis of steam and gas turbines. If a control volume enclosing the steam or gas turbine is properly selected, the difference in kinetic energy between inlet and exit ports is usually small enough to be neglected. The difference in potential energy is also typically negligible. Equation (5.26) then becomes:

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}(h_i - h_e) \quad (5.32)$$

The heat transfer between turbine and surroundings is usually small enough relative to the power and enthalpy difference terms, giving simply:

$$\dot{W}_{cv} = \dot{m}(h_i - h_e) \quad (5.33)$$

5.6 Compressors and Pumps

Compressors and **Pumps** are devices in which work is done on the substance flowing through them in order to change the state of the substance, typically to increase the pressure. The term *compressor* is used when the substance is a gas (vapor) and the term *pump* is used when the substance is a

liquid. Axial flow compressors are key component of turbojet engines. Compressors are also essential components of refrigeration and heat pump systems. Pumps are commonly used to fill water towers, remove water from flooded areas etc.

Figure 5.10 illustrates an axial flow compressor. Several other types of compressors are available including reciprocating, centrifugal, roots type etc. In schematics of thermal systems pumps and compressors are represented using the same symbol as a subsonic nozzle. The steady-state energy rate equation for these devices could be simplified in the same way as in turbines, yielding:

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} (h_i - h_e) \quad (5.34)$$

Heat transfer with the surroundings is frequently a secondary effect that can be neglected, giving

$$-\dot{W}_{cv} = \dot{m} (h_e - h_i) \quad (5.35)$$

For pumps, the kinetic and potential energy terms may be significant depending upon the application. Note that in Equation (5.35), $\dot{W}_{cv}$ is *negative* because a power *input* is required.

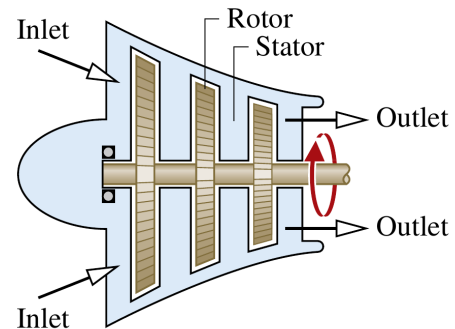


Figure 5.10: Schematic of an axial flow compressor.

Example Problem 03

Air enters a compressor operating at steady state at a pressure of 1 bar, a temperature of 290 K, and a velocity of 6 m/s through an inlet with an area of 0.1 m². At the exit, the pressure is 7 bar, the temperature is 450 K, and the velocity is 2 m/s. Heat transfer from the compressor to its surroundings occurs at a rate of 180 kJ/min. Employing the ideal gas model, calculate the power input to the compressor, in kW.

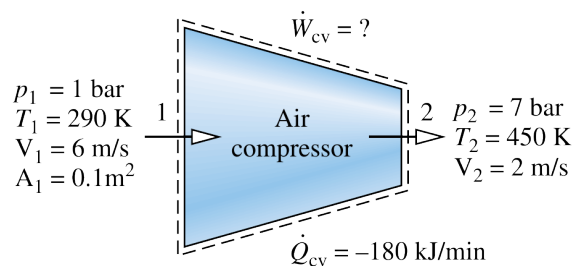
Solution

Steady-state mass rate balance for one inlet and one exit system is:

$$\dot{m} = \frac{A_1 V_1}{v_1} = \frac{A_2 V_2}{v_2} \quad (5.36)$$

The steady-state energy rate balance equation for one-inlet and one-exit system is:

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} \left[(h_i - h_e) + \frac{V_i^2 - V_e^2}{2} + g(z_i - z_e) \right] \quad (5.37)$$



Writing both the equations in the beginning helps us identify known and unknown quantities. This further helps in developing effective strategies to solve the problem. For ideal gas, two intensive properties are required to fully fix a state. Both temperature and pressure are provided at the inlet and exit of the control volume. Thus the states at stations 1 and 2 are *fixed*. However, since enthalpy and entropy for ideal gases are solely functions of temperature, only the temperature information will be required at the inlet and exit ports to compute enthalpy. At station 1, if the v_1 is computed from

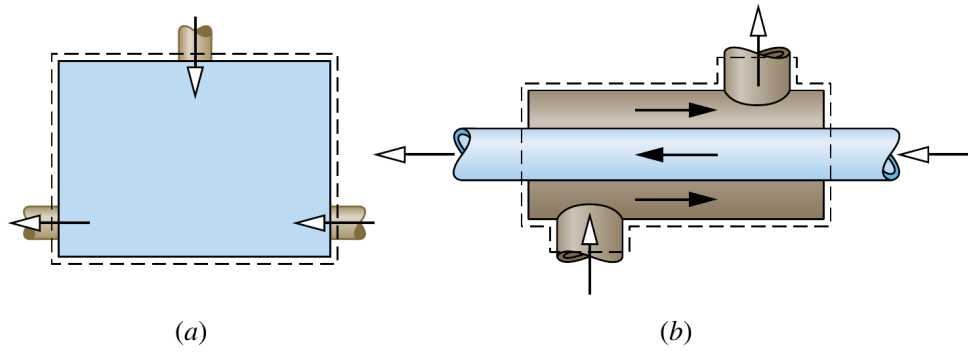


Figure 5.11: Common type of heat exchangers: (a) Direct contact type, (b) Surface contact type.

the ideal gas relation, we could find out the mass flow rate $\dot{m}$. The mass flow rate can be used in the energy rate balance equation to calculate power $\dot{W}_{cv}$.

Let us first compute v_1 from the ideal gas relation $Pv = RT$, where R is the gas constant (not the universal gas constant). $R = \hat{R}/\hat{M}$, where $\hat{R}$ and $\hat{M}$ are universal gas constant and molar mass, respectively. So:

$$v_1 = \frac{\hat{R}T_1}{\hat{M}p_1} = \frac{(8.314 \frac{\text{J}}{\text{mol}\cdot\text{K}})(290 \text{ K})}{(28.97 \times 10^{-3} \frac{\text{kg}}{\text{mol}})(10^5 \text{ Pa})} = 0.832 \frac{\text{m}^3}{\text{kg}} \quad (5.38)$$

The mass flow rate $\dot{m}$ can then be computed from Equation (5.36) as $\dot{m} = A_1 V_1/v_1 = 0.721 \text{ kg/s}$. Now to compute power input to the compressor, $-\dot{W}_{cv}$, we will employ Equation (5.37), after neglecting changes in potential energy:

$$\begin{aligned} -\dot{W}_{cv} &= -\dot{Q}_{cv} + \dot{m} \left[(h_2 - h_1) + \frac{V_2^2 - V_1^2}{2} \right] \\ &= - \left(-3000 \frac{\text{J}}{\text{s}} \right) + \left(0.721 \frac{\text{kg}}{\text{s}} \right) \left[(451.8 - 290.16) \times 10^3 \left(\frac{\text{J}}{\text{kg}} \right) + \frac{2^2 - 6^2}{2} \left(\frac{\text{J}}{\text{kg}} \right) \right] \\ &= \underset{\text{heat loss to surroundings}}{3 \text{ kJ}} + \underset{\text{enthalpy change}}{116.542 \text{ kJ}} - \underset{\text{kinetic energy change}}{0.012 \text{ kJ}} \\ &= 119.53 \text{ kJ} \end{aligned} \quad (5.39)$$

Equation (5.39) demonstrates that the heat loss to the surroundings and the contribution of kinetic energy change can indeed be neglected, with respect to the magnitude of enthalpy change.

5.7 Heat Exchangers

Heat exchangers have innumerable domestic and industrial applications, including use in home heating and cooling systems, automotive systems, electrical power generation, and chemical processing. There are two common type of heat exchangers available - (a) **direct contact**, and (b) **surface contact**. In direct contact heat exchangers, hot and cold streams are mixed directly. These type of heat exchangers are used in open feedwater heaters, desert coolers, evaporative cooling towers etc. An illustration is provided in Figure 5.11(a). Another common type of exchanger is one in which a gas or liquid is *separated* from another gas or liquid by a wall through which heat is conducted. Such type is commonly used in refrigeration systems, power plants, automobiles radiators, chemical processing etc. An illustration is provided in Figure 5.11(b). Heat exchangers can involve multiple inlets and exits. For a control volume enclosing a heat exchanger, the only work is the flow work at entry and

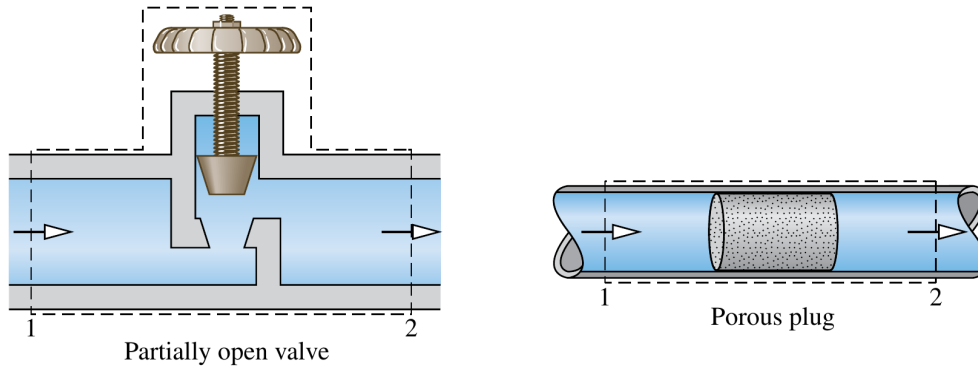


Figure 5.12: Examples of throttling devices.

exits, but this contribution is included in the enthalpy terms not $\dot{W}_{cv}$. In fact, $\dot{W}_{cv} = 0$ in the energy rate balance equation. In addition, the kinetic and potential energies of the flowing terms usually can be ignored at the inlets and exits. Therefore, the energy rate balance Equation (5.24) becomes:

$$0 = \dot{Q}_{cv} + \sum_i \dot{m}_i h_i - \sum_e \dot{m}_e h_e \quad (5.40)$$

5.8 Throttling Devices

A significant reduction in pressure can be achieved simply by introducing a restriction into a line through which a gas or liquid flows. This is commonly done by means of a partially opened valve or a porous plug. These *throttling* devices are illustrated in Figure 5.12. An application of throttling occurs in vapor-compression refrigeration systems, where a valve is used to reduce the pressure of the refrigerant from the pressure at the exit of the condenser to the lower pressure existing in the evaporator.

For a control volume enclosing a throttling device, the only work is flow work at locations where mass enters and exits the control volume, so the term $\dot{W}_{cv}$ drops out of the energy rate balance. There is usually no significant heat transfer with the surroundings, and the change in potential energy from inlet to exit is negligible. The energy rate balance Equation (5.24) then becomes:

$$0 = (h_1 - h_2) + \frac{V_1^2 - V_2^2}{2} \quad (5.41)$$

Although velocities may be high in the vicinity of the restriction imposed by the throttling device on the flow through it, measurements made upstream and downstream of the reduced flow area show in most cases that the change in the specific kinetic energy of the flowing substance between upstream and downstream locations can be neglected. With this further simplification, the above equation reduces to:

$$h_2 = h_1 \quad (5.42)$$

When the flow through the valve or restriction is idealized in this way, the process is called a throttling process.

Bibliography

- [1] Michael J Moran, Michael J Moran, Howard N Shapiro, Daisie D Boettner, and Margaret B Bailey. *Fundamentals of Engineering Thermodynamics*. 2018. ISBN: 978-1-119-39146-3 978-1-119-39147-0 978-1-119-39138-8.